ZIXUAN KE

Website ⋄ Github (≈1k stars) ⋄ Google Scholar (≈3k citations, 25 h-index) ⋄ zke4.uic@gmail.edu

BIOGRAPHY

I am a research scientist at Salesforce AI Research with 30+ papers (20+ first-authored) at top venues (NeurIPS, ICLR, etc.), with multiple oral presentations, a best paper nomination, and an outstanding survey award. My research builds **autonomous agentic systems** for an **ever-changing world** characterized by emerging domains, events, tools, skills, and agents. This spans three interconnected themes: (1) **Reasoning and Agentic Systems**: designing RL-based frameworks for agent orchestration, inference-time scaling, and multi-step reasoning; (2) **LLM Post-training**: adapting LLMs to new domains and tasks via RL, SFT, and continual pre-training while preserving general capabilities (1k+ citations); and (3) **Continual Learning**: enabling knowledge transfer and preventing catastrophic forgetting across tasks and domains (1k+ citations). My earlier work also includes **NLP** (sentiment analysis, dialogue) and **argument mining** (essay scoring, persuasiveness).

INDUSTRY EXPERIENCE

Salesforce AI Research, Research Scientist *May 2024 - present, Palo Alto, CA*

Agents, Reasoning and Post-training (RL, SFT, CPT). → 12 publications; 5 first-authored; 2 orals, 1 best paper nomination, 1 outstanding survey award

- Led **MAS-Orchestra** [ICML'26]: built RL training framework and function-calling abstraction for holistic multi-agent orchestration, composing entire MAS at each step rather than incrementally. Also built **MAS-Bench**, a controlled benchmark studying when MAS outperforms single-agent systems. Outperforms GPT-5 and Claude-Sonnet-4.5 by up to 23% across 5 benchmarks with 10× efficiency gain.
- Led **MAS-Zero** [SEA@NeurIPS'25, **Oral**]: built inference-time self-refinement framework for automatic MAS design with zero supervision, achieving Pareto-frontier performance across reasoning, coding, and agentic tasks (up to 16.7% accuracy gain).
- Led **LLM Reasoning Survey** [TMLR'25, **Outstanding Survey Award**; NeurIPS'25 Tutorial]: survey on inference scaling, learning to reason, and agentic systems.
- Led **FinDAP** [EMNLP'25, **Oral**, Best Paper Nomination]: built end-to-end post-training pipeline (CPT+SFT+DPO) for 7B–32B models, including synthetic data generation (FinTrain) and evaluation suite (FinEval), achieving SOTA across financial tasks. 5k+ HuggingFace downloads. Also delivered **NAACL'25 Tutorial** as sole speaker (3-hour tutorial on LLM post-training).
- Mentored 5 publications: **SkillOrchestra** [Preprint'26, #2 HuggingFace Daily]: skill-based agent routing outperforming SOTA RL orchestrators by 22.5% with 700× cost reduction; **MAS-ProVe** [ICML'26]: analysis of process verification for multi-agent systems; **SPARKLE** [NeurIPS'25]: analytical framework for RL in math reasoning; **LiveResearchBench** [ICLR'26]: benchmark for deep research requiring long-horizon reasoning and tool use, evaluating 17 frontier systems; **FaithEval** [ICLR'25]: 4.9K-problem contextual faithfulness benchmark.

Google DeepMind, Research Intern *Summer 2023, Mountain View, CA*

Retrieval-augmented Generation. Mentors: Weize Kong, Cheng Li, Mingyang Zhang and Qiaozhu Mei
Proposed preference-aligned retrieval for RAG, bridging the gap between retrievers and LLMs. → ACL'24

Meta AI, Research Intern *Summer 2022, Menlo Park, CA*

Continual Conversational Summarization. Mentors: Haoran Li, Wenhan Xiong and Asli Celikyilmaz
Developed sub-network discovery with soft-masking for continual learning of mixed tasks. → EMNLP'23

Amazon Science, Applied Scientist Intern *Summer 2021, Seattle, WA*

Multi-domain Imbalanced Learning. Mentors: Mohammad Kachuee and Sungjin Lee
Proposed domain-aware contrastive knowledge transfer for multi-domain imbalanced data. → Preprint

EDUCATION

Ph.D., University of Illinois at Chicago *Aug. 2019 - May 2024*
Ph.D. in Computer Science. GPA 4.0/4.0. Advisor: Bing Liu
LLM Post-training and Continual Learning

M.Sc., The University of Texas at Dallas *Aug. 2017 - Jun. 2019*
M.Sc. in Computer Science. Advisor: Vincent Ng
Argument Mining

B.Sc., South China Agricultural University *Aug. 2013 - Jun. 2017*
B.Sc. in Computer Science

SELECTED PUBLICATIONS

Full list on Google Scholar

Reasoning and Agentic Systems (Research Hub)

[1] **MAS-Orchestra: Understanding and Improving Multi-Agent Reasoning Through Holistic Orchestration and Controlled Benchmarks**
Zixuan Ke, Yifei Ming, Austin Xu, . . . , Semih Yavuz, Caiming Xiong, Shafiq Joty
*International Conference on Machine Learning **(ICML)***, 2026

[2] **MAS-Zero: Designing Multi-Agent Systems with Zero Supervision**
Zixuan Ke, Austin Xu, Yifei Ming, Xuan-Phi Nguyen, Caiming Xiong, Shafiq Joty
SEA@NeurIPS, 2025. Oral

[3] **A Survey of Frontiers in LLM Reasoning: Inference Scaling, Learning to Reason, and Agentic Systems**
Zixuan Ke, Fangkai Jiao, Yifei Ming, . . . , Silvio Savarese, Caiming Xiong, Shafiq Joty
NeurIPS Tutorial, 2025; ***TMLR***, 2025. Outstanding Survey Award

[4] **MAS-ProVe: Understanding the Process Verification of Multi-Agent Systems**
Vishal Venkataramani, Haizhou Shi, **Zixuan Ke**, . . . , Hao Wang, Shafiq Joty
*International Conference on Machine Learning **(ICML)***, 2026

[5] **SkillOrchestra: Learning to Route Agents via Skill Transfer**
Jiayu Wang, Yifei Ming, **Zixuan Ke**, Shafiq Joty
Preprint, 2026. #2 Paper of the day on huggingface

[6] **Dissecting Mathematical Reasoning for LLMs Under Reinforcement Learning**
Jiayu Wang, Yifei Ming, **Zixuan Ke**, Caiming Xiong, Shafiq Joty
*Advances in Neural Information Processing Systems **(NeurIPS)***, 2025

[7] **LiveResearchBench: A Live Benchmark for User-Centric Deep Research in the Wild**
Jiayu Wang, Yifei Ming, . . . , **Zixuan Ke**, Caiming Xiong, Shafiq Joty
*International Conference on Learning Representations **(ICLR)***, 2026

[8] **FaithEval: Can Your Language Model Stay Faithful to Context, Even If "The Moon is Made of Marshmallows"**
Yifei Ming, Shiva Purushwalkam, Sachin Pandit, **Zixuan Ke**, Caiming Xiong, Shafiq Joty
*International Conference on Learning Representations **(ICLR)***, 2025

LLM Post-training (Research Hub)

[9] **Demystifying Domain-adaptive Post-training for Financial LLMs**
Zixuan Ke, Yifei Ming, Xuan-Phi Nguyen, Caiming Xiong, Shafiq Joty
*Empirical Methods in Natural Language Processing **(EMNLP)***, 2025. Oral, Best Paper Nomination

[10] **Adaptation of Large Language Models**
Zixuan Ke, Yifei Ming, Shafiq Joty
NAACL 2025 Tutorial. Sole Speaker; 3-hour tutorial on LLM post-training

[11] **Bridging the Preference Gap between Retrievers and LLMs**
Zixuan Ke, Weize Kong, Cheng Li, Mingyang Zhang, Qiaozhu Mei, Michael Bendersky
*Association for Computational Linguistics (**ACL**)*, 2024

[12] **Continual Pre-training of Language Models**
Zixuan Ke*, Yijia Shao*, Haowei Lin* and Bing Liu
*International Conference on Learning Representations (**ICLR**)*, 2023

[13] **Adapting a Language Model While Preserving its General Knowledge**
Zixuan Ke, Yijia Shao, Haowei Lin, Hu Xu, Lei Shu and Bing Liu
*Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*, 2022a

[14] **Continual Training of Language Models for Few-Shot Learning**
Zixuan Ke, Haowei Lin, Yijia Shao, Hu Xu, Lei Shu and Bing Liu
*Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*, 2022b

Continual Learning

[15] **Continual and Contrastive Learning of Aspect Sentiment Classification Tasks**
Zixuan Ke, Bing Liu, Hu Xu and Lei Shu
*Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*, 2021

[16] **Continual Learning of a Mixed Sequence of Similar and Dissimilar Tasks**
Zixuan Ke, Bing Liu and Xingchang Huang
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2020

[17] **Achieving Forgetting Prevention and Knowledge Transfer in Continual Learning**
Zixuan Ke, Bing Liu, Nainzu Ma, Hu Xu and Lei Shu
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2021

[18] **Continual Learning of Natural Language Processing Tasks: A Survey**
Zixuan Ke and Bing Liu
Preprint, 2022

PROFESSIONAL ACTIVITIES

Invited Talks & Classes:

- LLM Reasoning and Agents
  Guest Lecture at Rice University, April 15, 2026
  Talk at CAMEL, Jan 23, 2026
- Adaptation of Large Language Models
  Tutorial at NAACL (Sole Speaker), May 3, 2025
  Talk at Visa Research, Mar 4, 2025
- Adapting Large Language Models for the Dynamic World
  Talk at Snowflake, Feb 1, 2024
  Talk at Salesforce AI Research, Jan 11, 2024
  Talk at Google DeepMind, Nov 9, 2023
- Continual Pre-training in Language Models , Talk at ContinualAI, Remote, April 27, 2023.
- Continual Learning in NLP. Tutorial at DEIM23, Remote, March 6, 2023
- Lifelong and Continual Learning. A Short PhD Course, Aalborg University, 2022.

Services:

- Area Chair: NeurIPS, ICML, ACL Rolling Review
- PC Member: ICLR, NeurIPS, ICML, ACL Rolling Review, TPAMI, TKDE, among others